MIL-F-18698(NOrd)

2 June 1955
SUPERSEDING
NAVORD OS 2958H
2 February 1955

MILITARY SPECIFICATION

FUZE, DETONATING, MARK 27 MOD 0 AND MOD 1 (40 MM.)

1. SCOPE

1.1 This specification governs the manufacture, loading, assembly, and preparation for delivery of Detonating Fuze Mk 27 Mod 0 and Mod 1 (40 mm.), and the methods of inspection and tests upon which acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-C-99	--Cement, Pettman.
JAN-P-108	--Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior.
JAN-C-388	--Compound, Sealing, N.R.C.
MIL-P-17141	--Plating; Protective, for Inert Fuze Parts.

STANDARDS

MIL-STD-105	--Sampling Procedure and Tables for Inspection by Attributes.
MIL-STD-129	--Marking for Shipment and Storage
MIL-STD-350	--Jolt Test for Use in Production of Fuzes.
MIL-STD-351	--Jumble Test for Use in Production of Fuzes.
MIL-STD-352	--Forty (40) Foot Drop Test for Use in Production of Fuzes.
MIL-STD-353	--Transportation Vibration Test for Use in Production of fuzes.

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DRAWINGS

BUREAU OF ORDNANCE

SK 90734	--40 mm. Detonating Fuze Mk 27 Mod 0 and Mk 27 Mod 1, List of Drawings.
SK 90735	--40 mm. Detonating Fuze Mk 27 Mod 0 and Mk 27 Mod 1, List of Drawings.
LD 109431	--Ammunition Component Box Mk 1.
LD 256003	--Ammunition Component Box Mk 2.
1386125	--Detonator Mk 18 Mod 0, Requirements and Testing Procedures.

And all documents listed on the above drawing, SK's and LD's.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	--General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by the contractor in accordance with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. --The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	--Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Officer, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 General.

3.1.1 The Detonating Fuze Mk 27 Mod 0 and Mod 1 shall be manufactured in accordance with List of Drawings SK 90734 and SK 90735 and all documents listed thereon.

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3.1.2 The requirements and procedures established in OP 400 shall be adhered to whenever applicable.

3.2 Interchangeability. --All parts shall be capable of interchangeable assembly without interference, binding, or misalignment. The contractor shall provide and use such jigs, gages, and gaging facilities as are required to insure compliance with this requirement.

3.3 Plating. --Plating shall conform to the requirements of Specification MIL-P-17141. Dimensions of plated parts shall apply after plating.

3.4 Detonator. --The detonator used shall be Detonator Mk 18 Mod 0, assembled and loaded as specified in Drawing 1386125.

3.5 Firing pin. --The firing pin shall have a sharp edge, 0.001 inch maximum radius, at the intersection of the flat end of the point and the conical surface and shall be inspected as specified in 4.4.1.

3.6 Loading and assembly. --All component parts shall be thoroughly cleaned and dried before assembly. Assembly shall proceed in a careful, neat and orderly manner so as to prevent the inclusion of dirt, oil, or foreign matter in the assembled fuze.

3.6.1 Booster lead-in assembly. --The magazine shall be placed in a suitable die and the booster lead-in hole loaded from the inside with a 5 second ram dwell time. The portion of tetryl broken off the funnel shall be approximately 0.020 inch long. The booster lead-in shall then be coated lightly with N.R.C. compound conforming to the requirements of Specification JAN-C-388 and allowed to dry. The minimum density of the lead-in shall be 1.44 grams per cubic centimeter.

3.6.1.1 The booster lead-in may be pre-pelleted and reconsolidated at the proper load after insertion into the magazine. Pressing shall be done from the cavity inside of the magazine.

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3.6.2 Magazine assembly. --Assembly of the booster pellet shall be accomplished by placing the dense or hard end toward the base of the fuze, away from the booster lead-in. The first pellet shall have a 0.01 inch minimum radius on the edge of its soft end next to the lead-in. The top end of the pellet shall be flush with the shoulder of the magazine. After the booster pellet has been assembled into the magazine, the magazine disc and magazine closing disc shall be immediately inserted and tightly crimped into place.

3.6.3 Final assembly. --Care shall be exercised to insure that the detonator is placed in the rotor so that the firing pin will strike the red sensitive priming end. Final seating of the detonator shall be accomplished by gently pressing until the rotor disc is held firmly between the open end of the detonator and the rotor and then staking the detonator in place.

3.6.3.1 It shall be insured that the rotor assembly is properly placed in the rotor block and that the rotor block assembly is properly placed in the fuze body.

3.6.3.2 After the threads of the magazine have been engaged one-half turn in the body, the remaining threads of the magazine shall be coated around the entire circumference with Pettman Cement conforming to the requirements of Specification JAN-C-99, and the magazine screwed into place. All excess cement shall then be removed. For proper waterproofing the density of the cement shall be maintained at the value indicated in Specification JAN-C-99.

3.7 Arming of subassemblies. --When inspected as specified in 4.4.2, the subassemblies shall arm within the rate of spin indicated in the applicable drawings.

3.8 Durability and safety. --The fuzes shall withstand the ruggedness tests specified in 4.5. The vibration test shall be conducted only when directed by the Government and normally will be restricted to preproduction samples and specially directed programs.

3.9 Non-Functioning. --Fuzes shall pass the flight test specified in 4.6.

3.10 Functioning. --Fuzes shall pass the functioning test specified in 4.7.

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3.11 Workmanship. --Only the best trade practice shall be utilized in the manufacture of the Detonating Fuze Mk 27 Mod 0 and Mod 1 (40 mm.). Care shall be exercised in the assembly of the component parts to assure proper seating and alignment especially where crimping or staking is employed to maintain parts in their relative positions. There shall be no burrs or other imperfections of manufacture harmfully affecting the functioning or assembly of the product. All parts shall be free of chips, dirt and other foreign materials. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent which is employed.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. --Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. --As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless othersie specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. --Before entering into quantity production, a preproduction sample of 250 Detonating Fuzes Mk 27 Mod 0 or Mk 27 Mod 1, as applicable, shall be delivered by the contractor for test to an agency designated by the Government. These fuzes shall be manufactured by the same procedures and processes and at the same location proposed by the contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the contractor will be notified and will be authorized by the Government to proceed with the submission of the product for acceptance. Accepted samples shall become the property of the Government and will be included in the number called for in the contract schedule.

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4.2.1 Preproduction sample for a subsequent contract. The necessity for a preproduction sample will be determined by the Government when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. --For procurement purposes each lot of detonating fuzes, of the Mod required, shall consist of 20,000 units plus those required for all test purposes. Unless otherwise stated in the contract, order of requisition, all units for test purposes shall be at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes as defined in 4.1.1.

4.4 Inspection. --The general inspection procedures shall be in accordance with General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OP 400; Government inspection shall be in accordance with the provisions of General Procedure for Government Inspection and Contractor's inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.4.1 Firing pin. --The points of the firing pin shall be inspected with a shadowgraph comparator. Points which show a radius in excess of 0.001 inch shall be rejected.

4.4.2 Spin test inspection. --Subassemblies containing the firing pin detents, when applicable, and subassemblies containing the rotor detents and the rotor shall be given a spin test inspection. The rate of spin at which the subassemblies arm shall be in accordance with the applicable drawings.

4.4.2.1 This inspection shall be conducted at the inert parts manufacturing plant and at the loading activity.

4.5 Durability and safety tests.

4.5.1 Jolt. --The test shall be conducted on the basis of 2 completely live loaded fuzes selected from each production lot. The fuzes shall be tested in accordance with Standard MIL-STD-350. The testing machine shall be equipped with suitable adapters to receive the fuzes. Failure of any fuze to pass this test shall cause rejection of the lot.

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4.5.2 Jumble. --Two completely live loaded fuzes selected from each production lot shall be subjected to a jumble test in accordance with the provisions of Standard MIL-STD-351. Appropriate jumbling fixtures shall be used. Failure of any fuze to withstand this test shall be cause for rejection of the lot.

4.5.3 Hard surface drop test. --Three completely live loaded fuzes, except for inert boosters, shall be selected from each production lot. These fuzes shall be installed by a suitable means in three 40 mm. complete rounds which are inert loaded to weight, and subjected to a 40 foot drop on armor plate. One round shall be dropped on its nose, one on its side and the third one on its base. The precautionary and maintenance provisions of Standard MIL-STD-352 shall be applicable to the test. Functioning of any fuze on strking the plate or derangement of parts which may cause functioning shall cause rejection of the lot.

4.5.4 Vibration. --The test shall be conducted on the basis of 20 fuzes and as prescribed by Standard MIL-STD-353.

4.6 Flight (non-functioning) test. --Two completely live loaded fuzes from each production lot shall be assembled in service-loaded (without self-destroying feature) projectiles, and the projectiles fired from a gun. The fuzes shall not function to explode the projectiles in the gun or during flight. This test may be discontinued at the discretion of the Government.

4.7 Functioning test. --Select and test completely live loaded fuzes from each production lot in accordance with the Single Sample Plan, AQL 1.5 percent, Table I of Standard MIL-STD-105.

4.7.1 The fuzes shall be assembled in live loaded 40 mm. complete rounds whcih shall be fired at service velocity and 0 degree obliquity against an approved 0.138 inch thick chipboard target placed at a distance of 100 to 400 feet from the gun. Each fuze tested shall cause a high order detonation of the projectile within two feet after contact with the target.

4.7.2 Reduced inspection may be instituted provided the conditions prescribed by Standard MIL-STD-105 are satisfied.

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4.7.3 The occurrence of a premature attributable to a fuze shall cause rejection of the lot. Subsequent lots shall be tested in accordance with the sampling plan specified in 4.7.

4.8 Surveillance. --Ten fuzes from each production lot shall be forwarded to a depot designated by the Government for retention as a surveillance sample. The surveillance fuzes shall be plainly tagged:

Surveillance Sample Fuze Mark 27 Mod _____
Lot_____ Date of Loading _____
Inert Mfr. _____ Loaded by _____

Retain at designated ammunition depot for use in surveillance test as directed by the Government.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. --One hundred and fifty (150) fuzes shall be packaged and packed in Ammunition Component Box Mk 1 or Mk 2 in accordance with LD 109431 or LD 256003 respectively. Partitions and separators shall conform to Specification JAN-P-108, W5G.

5.2 Marking. --In addition to any special marking required by the contract or order, all containers shall be marked in compliance with Standard MIL-STD-129 and the Code of Federal Regulations 49 CFR 71-78.

6. NOTES

6.1 Intended use. --The Detonating Fuze Mk 27 Mod 0 and Mod 1 are intended for use with 40 mm. projectiles.

6.2 Ordering data. --Procurement documents should specifiy the following:

(a) Title, symbol and date of this specification
(b) Whether Detonating Fuze Mk 27 Mod 0 or Mk 27 Mod 1 is reuquired (see 1.1).
(c) Whether or not vibration test will be required (see 3.8).

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Patent notice. --When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-18698(NOrd)	Fuze, Detonating, Mark 27 Mod 0 and Mod 1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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